ENMA 300 / ENME 382: Solving Guide for an AI Assistant

Introduction to Engineering Materials, UMD, Fall 2026. Textbook: Callister & Rethwisch, *Materials Science and Engineering: An Introduction*. Covers Chapters 1 to 6: bonding, crystal structures, ceramics, polymers, defects, diffusion.

> **Instructions to the AI.** Use this guide to solve homework and exam-style questions for this course. The student's exam allows only a one-page, front-and-back formula sheet, which is attached at the end of this document. Treat that sheet, plus any data given in the question itself (radii, atomic weights, D_0, Q_d, erf tables), as the only reference available. Solve with those formulas and values, show the setup and each step, convert to SI before plugging in, and give the final answer to 3 significant figures unless the question asks otherwise. Answer exactly what is asked, in the format the input box expects.

1. Answer conventions that matter for grading

- **Significant figures:** WileyPlus graded 35,646 wrong and 35,600 right. Report 3 significant figures (e.g. 3.56×10^{4}, 1.02×10^{-15}).
- **Number entry:** scientific notation as `1.53e-15`, no commas, no units in the box. "Non-scientific" means write it out in full (0.0000000126).
- **Given constants win:** if the problem lists atomic weights (e.g. H = 1.008), use exactly those. The textbook example uses H = 1.01, which shifts the last digit.
- **Randomized values:** WileyPlus can change numbers between attempts. Always re-read the current values before recomputing.
- **Fractions as decimals** when asked ("represent fractional indices as decimals": 0.5, not 1/2).
- **Notation:** planes (hkl), plane families {hkl}, directions [uvw], direction families <uvw>. Negative index = bar over the number, or a minus sign in text. No commas inside brackets.
- **Written homework style** (Gradescope): plain text, Times New Roman, no bold, symbols typed as D_0, Q_d, 10^-7, sqrt(). The student prefers short direct answers.

2. Bonding

Key formulas

- Percent ionic character: $\%IC = \{1 - \exp[-0.25(X_A - X_B)^2]\} \times 100$, X = Pauling electronegativity.
- Coulomb attraction: $F_A = (|Z_1|e)(|Z_2|e) / (4\pi\varepsilon_0 r^2)$, $1/(4\pi\varepsilon_0) = 8.99\times10^{9}$ N·m²/C², e = 1.602×10^{-19} C. At equilibrium (ions touching) $r = r_{cation} + r_{anion}$ and $F_R = F_A$ in magnitude, because net force is zero.
- Rearranged: $r = \sqrt{(kZ_1Z_2e^2/F_A)}$, then $r_{anion} = r - r_{cation}$.

Element	X	Element	X	Element	X	Element	X
Cs	0.7	Al	1.5	Fe	1.8	N	3.0
K	0.8	Ti	1.5	Ni	1.8	Cl	3.0
Na	0.9	Ga	1.6	Si	1.8	O	3.5
Ca	1.0	Zn	1.6	Cu	1.9	F	4.0
Mg	1.2	Cr	1.6	As	2.0	Br	2.8
H	2.1	C	2.5	S	2.5	Pb, Sn	1.8

Values from Callister Fig. 2.9. If the course figure differs, use the course figure.

Worked examples (confirmed in this course)

Divalent cation 0.065 nm, monovalent anion 0.126 nm. r = 0.191 nm. $F_A = (8.99\times10^{9})(2)(1)(1.602\times10^{-19})^2 / (1.91\times10^{-10})^2 =$ **1.26×10^{-8} N**; $F_R =$ **1.26×10^{-8} N**.

$F_A = 1.60\times10^{-8}$ N, both divalent, $r_{cation} = 0.069$ nm. $r = \sqrt{[(8.99\times10^{9})(4)(1.602\times10^{-19})^2/1.60\times10^{-8}]} = 0.240$ nm → **$r_{anion} = 0.171$ nm**.

%IC: CsBr (0.7, 2.8) **66.8%** · GaAs (1.6, 2.0) **3.9%** · Fe_3C (1.8, 2.5) **11.5%** · NaF (0.9, 4.0) **91.0%** · Si_3N_4 (1.8, 3.0) **30.2%**

Concept answers

Question	Answer
Inert gas configuration	Filled outer shell, ends $1s^2$ (He) or ns^2np^6 (e.g. $1s^22s^22p^63s^23p^6$ = Ar)
Brass (Cu–Zn), steel, titanium	Metallic
Rubber, polyester, epoxy, other polymers	Covalent with some van der Waals
Solid krypton	van der Waals
Al_2O_3	Ionic (some covalent), non-directional
Si, SiC	Covalent, directional (tetrahedral)
Why is ionic non-directional?	Coulomb force depends only on distance, not direction
Why covalent is directional	Shared electron pairs sit in specific orbitals at fixed bond angles
Why APF of SiC (zinc blende) ≪ Cu (FCC 0.74)	Directional covalent bonds lock atoms at fixed angles, so packing is loose; metallic bonds are non-directional so atoms pack densely
Thermal conductivity Pt ≫ Al_2O_3 ≫ HDPE	Metals conduct with free electrons; ceramics only with phonons through a rigid ordered lattice; polymers only with phonons across weak vdW links between tangled chains, which scatter phonons
Same 304 steel, cold worked vs annealed: same CTE?	Yes. CTE depends on interatomic bonding (composition, crystal structure), not processing or microstructure
Why ceramics, glass, Si, epoxy list no yield strength	Brittle: no mechanism for plastic deformation (no mobile dislocations, rigid network), so they fracture before yielding
Why Al_2O_3/ZrO_2 strength ≈ metals but metals used structurally	Strong ionic/covalent bonds give high strength, but ceramics are flaw-sensitive and brittle; metals plastically deform and blunt cracks
304 stainless	~18–20% Cr, 8–10.5% Ni, ≤0.08% C, ≤2% Mn, ≤1% Si, balance Fe; austenitic FCC; kitchen, food, chemical equipment

Potential energy (E–r) curve rules

Curve feature	Property
Deeper well (larger E_0)	Higher melting point, stronger bonds
Narrow well, high curvature at r_0; steeper force-curve slope at r_0	Higher elastic modulus (stiffer)
Deep, narrow, symmetric well	Lower coefficient of thermal expansion
Shallow, wide, asymmetric well	Higher CTE

For a force curve, look at the slope at r_0, not the height. For an energy curve, depth = bond energy. Example from homework: B deepest and narrowest → B > A > C for T_m and E; CTE C > A > B. To join two materials (bilayer), pick the pair with the most similar CTE and E.

3. Crystal structures

Structure	a in terms of R	Atoms/cell n	CN	APF
Simple cubic	a = 2R	1	6	0.52
BCC	a = 4R/√3	2	8	0.68
FCC	a = 2R√2	4	12	0.74
HCP	a = 2R, c = 1.633a	6	12	0.74

- Density: $\rho = nA/(V_C N_A)$. Identify structure: $n = \rho V_C N_A/A$ and round, or compute ρ for each candidate and match.
- **Hexagonal cell volume** = full hexagonal prism = $(3\sqrt{3}/2)a^2c$. Using $a^2c \cdot \sin 120°$ (the rhombic primitive cell, 1/3 the size) was graded wrong.
- Seven systems: cubic a=b=c, 90°; tetragonal a=b≠c, 90°; hexagonal a=b≠c, 90/90/120°; rhombohedral a=b=c, equal angles ≠90°; orthorhombic a≠b≠c, 90°; monoclinic a≠b≠c, α=γ=90°≠β; triclinic all unequal.
- a = b = c: cubic and rhombohedral. **a = b ≠ c: tetragonal AND hexagonal** (answer "Both B and G" was correct; tetragonal alone was wrong).
- Metals, ceramics and polymers can all form crystalline solids ("all of these").

Au, R = 0.144 nm (FCC): a = 2√2(0.144) = 0.4073 nm, V = **6.76×10^{-29} m³**.
Ag FCC, ρ = 10.5, A = 107.87: $V_C = 4(107.87)/(10.5\cdot6.022\times10^{23}) = 6.825\times10^{-23}$ cm³, a = 4.086×10^{-8} cm, R = a√2/4 = **1.445×10^{-8} cm**.
BCC, R = 0.150 nm, A = 70.5: a = 0.3464 nm, $\rho = 2(70.5)/[(3.464\times10^{-8})^3\cdot N_A]$ = **5.63 g/cm³**.
Cubic, a = 0.288 nm, ρ = 7.20, A = 52.0: n ≈ 2 → BCC → **APF 0.68**.

Points, directions, planes

- **Point coordinates:** fractions of a, b, c along x, y, z from the origin corner. Read the figure's axes carefully (in the NaCl figure, the atoms on the back face had y = 1, not 0).
- **Direction:** head − tail, then multiply by a common factor to get the smallest integers. An endpoint at an edge midpoint gives a ½, which becomes a 2. Example: tail (1, 0, 0), head (0, 1, ½) → (−1, 1, ½) → [−2 2 1]. Vector with positive a and b and negative c components of equal size → [1 1 −1].
- **Plane:** if the plane passes through the origin, move the origin to a nearby corner so the plane doesn't touch it. Then intercepts (∞ if parallel), reciprocals, clear fractions. Example: intercepts 1/2, ∞, 1/3 → (203). Intercepts 1/2, 2/3, −2/3 → 2, 3/2, −3/2 → (4 3 −3).
- Intersection line of two planes in a cubic system = cross product of their normals: (110) × (11−1) = [−1 1 0].
- Choosing a different valid origin can give the negative of the same indices; both describe the same plane.

4. Ceramics

r_C/r_A	CN	Geometry
< 0.155	2	linear
0.155–0.225	3	triangular
0.225–0.414	4	tetrahedral
0.414–0.732	6	octahedral
0.732–1.0	8	cubic

Structure	Formula	CN (cation/anion)	Anion packing	Formula units n'
Rock salt (NaCl)	AX	6/6	FCC	4
Cesium chloride	AX	8/8	simple cubic	1
Zinc blende	AX	4/4	FCC	4
Fluorite (CaF_2)	AX_2	8/4	simple cubic	4
Antifluorite	A_2X	4/8	FCC	4
Perovskite	ABX_3	12 (A), 6 (B)	FCC	1

- $\rho = n'(\Sigma A_C + \Sigma A_A)/(V_C N_A)$, so $n' = \rho V_C N_A/\Sigma A$ (use this to pick between structures).
- Rock salt: cation and anion touch along the edge, $a = 2(r_C + r_A)$.
- Exams mostly test AX types: compute the radius ratio, read the CN, match the structure.
- Two characteristics that determine ceramic structure: magnitude of the charge on each ion, and relative sizes of cation and anion.
- Schottky defect in Al_2O_3: 2 Al^{3+} vacancies + 3 O^{2-} vacancies, 0 interstitials.
- A_2X with $r_A = 0.125$, $r_X = 0.145$: CN of A = 4. Variant of the **fluorite** structure; X ions at **all** cube corners; **half of the** cube centers occupied by A.

AX: r_A 0.09, r_X 0.135 → 0.667 → CN 6 → **sodium chloride**.
AX, ρ = 3.35, a = 0.40 nm, A = 90.5 + 37.3: $n' = 3.35(0.40\times10^{-7})^3(6.022\times10^{23})/127.8 \approx 1$ → **cesium chloride**.
MgO (rock salt, r 0.072/0.140): a = 0.424 nm, $\rho = 4(40.31)/(a^3 N_A)$ = **3.51 g/cm³** (1.9% from 3.58), APF = $(4/3)\pi\cdot4(0.072^3 + 0.140^3)/0.424^3$ = **0.685**.
Cr_2O_3 hexagonal, a = 0.4961, c = 1.360 nm, ρ = 5.22: $V_C = 2.598a^2c$ = **8.70×10^{-22} cm³**; $n' = 5.22\cdot V_C\cdot N_A/152.0$ = **18**; $V_S = 36\cdot(4/3)\pi(0.062)^3 + 54\cdot(4/3)\pi(0.140)^3$ = **6.57×10^{-22} cm³**; APF = **0.755**.

5. Polymers

Polymer	Repeat unit	m (g/mol)	Identifying feature
Polyethylene	C_2H_4	28.05	all H
Polypropylene	C_3H_6	42.08	one CH_3 side group
PVC	C_2H_3Cl	62.50	one Cl
PTFE	C_2F_4	100.02	all F
Polystyrene	C_8H_8	104.14	C_6H_5 (benzene) side group
Poly(vinyl alcohol)	C_2H_4O	44.05	one OH
PMMA	$C_5H_8O_2$	100.12	CH_3 + $COOCH_3$

- $DP = \bar{M}_n/m$. $\bar{M}_n = \Sigma x_i M_i$ (number fractions), $\bar{M}_w = \Sigma w_i M_i$ (weight fractions), M_i = midpoint of each range.
- Possible polymer for given distribution + DP: $m = \bar{M}_n/DP$, compare with PE 28.05, PTFE 100.02, PVC 62.50. Example: $\bar{M}_n$ = 44,600, DP 422 → m = 105.7 → none of them.
- Alternating poly(isobutylene-isoprene) repeat unit: $–CH_2–C(CH_3)_2–$ joined to $–CH_2–C(CH_3)=CH–CH_2–$.
- Linear polymers: PE, PP, PS (one double bond per monomer). Phenol-formaldehyde is a network (three reactive sites per phenol).
- % crystallinity = $\rho_c(\rho_s - \rho_a)/[\rho_s(\rho_c - \rho_a)]\times100$. From two samples (C as fractions): $\rho_c = \rho_{s1}\rho_{s2}(C_2-C_1)/[\rho_{s1}(1-C_1) - \rho_{s2}(1-C_2)]$, $\rho_a = \rho_{s1}\rho_{s2}(C_1-C_2)/(C_1\rho_{s1} - C_2\rho_{s2})$. Then plug the new ρ_s into the first equation.
- Number of chains in a part: total mass (ρ × volume) ÷ chain mass (DP × m) gives moles of chains; × N_A.

PP, $\bar{M}_n$ = 1,500,000: DP = 1,500,000/42.08 = 35,646 → entered **35,600** (accepted).
PE: m = **28.05**; DP = 24,000 → $\bar{M}_n$ = 24,000 × 28.05 ≈ **673,000**.
PP distribution (midpoints 12k–52k): $\bar{M}_n$ = **32,900**, $\bar{M}_w$ = **36,400**, DP = 32,880/42.08 = **781**.

Concept	Answer
Crystalline vs amorphous density, same polymer	Crystalline > amorphous (tighter chain packing)
Amorphous polymer optical behavior	Transparent (no crystalline/amorphous boundaries to scatter light)
Atactic PVC vs isotactic PP	Isotactic PP crystallizes more
Syndiotactic PP vs crosslinked cis-polyisoprene	Syndiotactic PP
Network phenol-formaldehyde vs isotactic PS	Isotactic PS
Block vs graft copolymer	Not possible to determine
Heated polymer chars instead of melting	Thermoset (crosslinked)
Why properties come as ranges	Molecular weight and distribution, % crystallinity, additives, copolymer ratio/tacticity/crosslink density, processing and test conditions
Recyclable?	Thermoplastics yes (PVC, PP); vulcanized elastomers (NBR) and thermosets (epoxy) no
Glass transition T_g	Temperature where amorphous regions change from brittle/glassy (below) to rubbery (above). Only for amorphous material; metals have none

6. Defects and composition

- Equilibrium vacancies: $N_v = N \exp(-Q_v/kT)$, $N = N_A\rho/A$. k = 8.62×10^{-5} eV/atom·K with Q in eV; use R with Q in J/mol. Fraction = N_v/N.
- Vacancy fraction from given N_v: $N = \rho N_A/A$ (convert cm^{-3} to m^{-3} by $\times10^6$), fraction = N_v/N. Example: $N_v = 0.5\times10^{25}$ m^{-3}, ρ = 7.40, A = 85.5 → $N = 5.21\times10^{28}$ → **9.59×10^{-5}**.
- wt% → at%: $C'_i = (C_i/A_i)/\Sigma(C_j/A_j)\times100$. Example 7.5 wt% Pb / 92.5 Sn → **4.44 at% Pb, 95.6 at% Sn**.
- at% → wt%: $C_i = C'_iA_i/\Sigma(C'_jA_j)\times100$. Example 44.8 Ag / 46.4 Au / 8.8 Cu at% → **33.3 / 62.9 / 3.85 wt%**. 5.5 at% Pb → 9.22 wt% Pb.
- kg per m^3: $C''_1 = C_1/(C_1/\rho_1 + C_2/\rho_2)\times10^3$. Example 0.55 wt% Si in Fe → **42.7 kg/m^3**.
- Hume-Rothery (complete solid solubility): radius difference < 15%, same crystal structure, similar electronegativity, same valence. Example: of Mg–Zn, Al–Zr, Cr–V, only **Cr–V** qualifies.
- Interstitial site sizes (r in terms of host R): SC face center 0.414R, body center 0.732R; FCC octahedral 0.414R, tetrahedral 0.225R; BCC octahedral 0.155R, tetrahedral 0.291R.
- Schottky: cation and anion vacancies in the formula's charge-neutral ratio. Frenkel: a cation moves into a nearby interstitial site. A compound needing more vacancies per defect ($MgCl_2$ needs 3, NaCl needs 2) and stronger-bound ions forms fewer Schottky defects.
- Dislocations: edge b ⊥ line, screw b ∥ line, mixed neither (most real dislocations are mixed). Dislocations barely move in ceramics, so they are brittle.
- Surface energy: denser planes have fewer broken bonds, so lower energy. BCC: γ(100) > γ(110). FCC densest plane (111).

- Metals vs ceramics vs polymers: intrinsic equilibrium point defects → metals/ceramics; impurities/dopants, dislocations, external surfaces → both; grain size → metals/ceramics; % crystallinity → polymers.

7. Diffusion

- Fick's 1st law (steady state): $J = -D\,dC/dx = -D(C_B - C_A)/(x_B - x_A)$; $J = M/(At)$.
- Fick's 2nd law, constant surface concentration: $(C_x - C_0)/(C_s - C_0) = 1 - \text{erf}[x/(2\sqrt{(Dt)})]$. C_0 initial, C_s surface, C_x at depth x.
- Same composition: $x^2/(Dt)$ is constant. At one temperature $x_1^2/t_1 = x_2^2/t_2$; at one depth $D_1t_1 = D_2t_2$.
- $D = D_0\exp(-Q_d/RT)$. Solve for T: $T = Q_d/[R(\ln D_0 - \ln D)]$. Two points: $Q_d = -R\ln(D_2/D_1)/(1/T_2 - 1/T_1)$, $D_0 = D_1\exp(Q_d/RT_1)$.
- log D vs 1/T plot: slope $= -Q_d/(2.3R)$. Steeper line = larger Q_d.
- Linear interpolation: $z = z_1 + (\text{erf} - \text{erf}_1)(z_2 - z_1)/(\text{erf}_2 - \text{erf}_1)$. The exam requires interpolation; the erf table is provided.
- Thin film (fixed amount β, not replenished): $C = [\beta/(2\sqrt{(\pi Dt)})]\exp(-x^2/4Dt)$. Steel carburizing uses the constant-C_s (erf) case; semiconductor dopant drive-in uses the thin-film case.
- Diffusion is faster for interstitial vs vacancy mechanisms (no vacancy-formation energy, many open sites; C in Fe ≫ Fe self-diffusion), along grain boundaries and dislocations (open structure, $Q \approx \frac{1}{2}Q_{bulk}$, dominate at low T while bulk dominates at high T), in open vs close-packed structures, for cations vs anions (smaller), and in secondary-bonded vs covalent materials.

Problem	Setup	Answer
N_2 through 3.6 mm steel, D 5.8×10^{-11}, J 2.9×10^{-7}, C 4.9 → 2.7 kg/m³	$x = D(C_A - C)/J$	4.40×10^{-4} m
D at 567°C, D_0 5.9×10^{-6}, Q 157 kJ/mol	T = 840 K	1.02×10^{-15} m²/s
BCC Fe 2.1 mm, 0.011 / 0.0070 wt% C, J 1.7×10^{-8}	C'' = 0.865, 0.551 kg/m³; $D = J\Delta x/\Delta C$	1.13×10^{-10} m²/s
11.1 h for 1.8 mm → time for 4.4 mm	$t_2 = t_1(x_2/x_1)^2$	66.3 h
Cr in Ni, D_0 1.1×10^{-4}, Q 272 kJ, D 1.3×10^{-14}	$T = Q/[R\ln(D_0/D)]$	1158°C
11 mm plate, 0.86/0.40 kg/m³, J 6.6×10^{-10}, D_0 6.2×10^{-7}, Q 80 kJ	$D = 1.58\times10^{-11}$	910 K
Fe in Cr, log D vs 1/T graph	read $(0.60\times10^{-3}, 1.87\times10^{-15})$, $(0.65\times10^{-3}, 2.26\times10^{-16})$	$Q \approx 3.5\times10^5$ J/mol, $D_0 \approx 1.9\times10^{-4}$
H_2 through 4 mm Pd, 0.25 m², D 6.0×10^{-8}, 5.8/0.21 kg/m³	J·A·3600	0.0755 kg/h
N in α-Fe 750°C, D_0 4.7×10^{-7}, Q 76.6 kJ, 1.5 mm, 3.2/0.40 kg/m³, 0.20 m², 12 h	D, J, M = JAt	D 5.76×10^{-11}; J 1.08×10^{-7}; 0.930 g
Carburize γ-Fe 1000°C, C_0 0.15, C_s 1.10, C_x 0.45 wt% at 1.2 mm	erf(z) 0.6842, z 0.7096	D 1.94×10^{-11}; t 10.2 h; at 1075°C 4.63 h (54.7% less)
Cu in Si: 7.80×10^{-11} at 573 K, 1.57×10^{-10} at 623 K	two-point method	Q 41.5 kJ/mol = 0.430 eV/atom; D_0 4.76×10^{-7}

Unresolved: Ni in Cu at 810°C with Table 6.2 values (1.9×10^{-4}, 230,000) computes to 1.53×10^{-15} m²/s but was marked wrong. Re-check the current temperature in the problem before reusing.

8. Material selection (guaranteed on the exam)

1. Read the service conditions: temperature range, loads, weight limits, environment, electrical needs, cost.
2. Rank the most important properties (e.g. CTE and temperature range first, density second for a launched part).

3. Eliminate candidates that fail (polymers whose T_g or T_m falls in the service range, materials too dense, too brittle, or conductive where insulation matters).
4. Compare 2 or 3 finalists with advantages and drawbacks.
5. Defend one choice and say how to mitigate its main drawback.

> Satellite instrument panel launched by rocket: choose an **aluminum alloy**. Low density (less rocket mass), low cost, low CTE, high yield strength, no T_g so it tolerates large temperature swings in orbit. Drawback: electrically conductive, so add insulating coatings, standoffs, potting, and careful wire routing. Dense alloys (e.g. steels) are poor because of launch mass; polymers like UHMWPE or PVC risk property loss near T_g.

There is no single right answer; any well-supported argument earns credit.

9. Exam logistics (from the review session)

- One self-made sheet, front and back, anything on it, printed is fine.
- Provided on the exam: an equation sheet, atomic weights or a periodic table, polymer repeat-unit structures, D_0 and Q_d values, the erf table, R and k.
- Most common mistake: units. Convert everything to SI first.

Appendix: the student's formula sheet

The next two pages are the formula sheet the student will have on the exam. When answering exam-style questions, rely on it plus the data in the question.

Constants & units

$N_A = 6.022\times10^{23}$ /mol · R = 8.314 J/mol·K · k = 1.38×10^{-23} J/atom·K = 8.62×10^{-5} eV/atom·K

e = 1.602×10^{-19} C · 1 eV = 1.602×10^{-19} J · $1/(4\pi\varepsilon_0) = 8.99\times10^9$ N·m²/C² ($\varepsilon_0 = 8.85\times10^{-12}$)

eV/atom × $(1.602\times10^{-19})(6.022\times10^{23})$ = J/mol · 1 eV/atom = 96,500 J/mol

1 nm = 10^{-9} m = 10^{-7} cm · 1 nm³ = 10^{-21} cm³ · 1 Å = 0.1 nm · 1 g/cm³ = 1000 kg/m³ · K = °C + 273

Convert to SI first. WileyPlus usually wants 3 sig figs.

Atomic weights (g/mol)

H 1.008 · C 12.01 · N 14.01 · O 16.00 · F 19.00 · Na 22.99 · Mg 24.31 · Al 26.98 · Si 28.09 · Cl 35.45 · Ti 47.87 · Cr 52.00 · Fe 55.85 · Ni 58.69 · Cu 63.55 · Zn 65.38 · Ag 107.87 · Sn 118.71 · Au 196.97 · Pb 207.2

Bonding

%IC = $\{1 - \exp[-0.25(X_A - X_B)^2]\} \times 100$

Coulomb: $F_A = (|Z_1|e)(|Z_2|e)/(4\pi\varepsilon_0 r^2)$, $r = r_{cation} + r_{anion}$. At r_0: $F_R = F_A$ (net F = 0). Solve for r: $r = \sqrt{(kZ_1Z_2e^2/F)}$

$E_N = -A/r + B/r^n$; F = dE/dr; at r_0: dE/dr = 0, E_0 = bond energy

E–r curve feature	Property
Deeper well (larger E_0)	Higher T_m, stronger bond
Narrow / high curvature at r_0 (steep F slope)	Higher E (modulus)
Deep, symmetric well	Lower CTE (α)
Shallow, wide, asymmetric	Higher CTE

Bond	Dir?	Where
Ionic (Coulomb)	Non	metal + nonmetal, ceramics
Covalent (shared e^-)	Dir	Si, C, polymer chains, SiC
Metallic (e^- sea)	Non	metals, alloys (brass)
van der Waals	weak	between polymer chains, noble gas solids (Kr)

Polymers/rubber/epoxy = covalent + vdW. Noble gas config ends ns^2np^6.

Metals: ductile, conductive, dense, opaque. **Ceramics:** ionic/covalent, stiff, brittle, insulating, low toughness, no yield (fracture first). **Polymers:** low ρ, low E, insulating.

Thermal k: metal (free e^-) > ceramic (phonons, rigid lattice) > polymer (weak vdW, scattering). CTE & E depend on bonding/composition, NOT processing.

Unit cells

	a vs R	n	CN	APF
SC	a = 2R	1	6	0.52
BCC	a = 4R/√3	2	8	0.68
FCC	a = 2R√2	4	12	0.74
HCP	a = 2R, c = 1.633a	6	12	0.74

$\rho = nA/(V_C N_A)$ · $V_C = a^3$ (cubic) · **HCP/hex $V_C = (3\sqrt{3}/2)a^2c = 2.598a^2c$**

APF = $V_{spheres}/V_C$, $V_{sphere} = (4/3)\pi R^3$

Unknown structure: $n = \rho V_C N_A/A$ → round (1 SC, 2 BCC, 4 FCC). Or compute ρ for BCC and FCC, see which matches.

Radius from ρ (FCC): $V_C = nA/(\rho N_A)$, $a = V_C^{1/3}$, $R = a\sqrt{2}/4$

LD = atoms centered on line / length · PD = atoms centered on plane / area. Densest: FCC (111), BCC (110). Denser plane = lower surface energy.

System	Edges	Angles
Cubic	a=b=c	all 90°
Tetragonal	a=b≠c	all 90°
Hexagonal	a=b≠c	90, 90, 120°
Rhombohedral	a=b=c	all equal ≠90°
Orthorhombic	a≠b≠c	all 90°
Monoclinic	a≠b≠c	α=γ=90°≠β
Triclinic	a≠b≠c	all ≠, ≠90°

a=b=c → cubic + rhombohedral. a=b≠c → tetragonal + hexagonal.

Points, directions, planes

Point: q r s = fractions of a, b, c (no brackets, use decimals if asked).

Direction [uvw]: head − tail → multiply to smallest integers → [ ]. Negative = bar. Family <uvw>. e.g. tail (1,0,0), head (0,1,1/2): (−1, 1, 1/2) ×2 → [−2 2 1]. Midpoint of an edge gives a 1/2 → index 2.

Plane (hkl): if plane hits origin, MOVE origin to a corner off the plane → intercepts (∞ if parallel) → reciprocals (1/∞ = 0) → clear fractions → ( ), no commas. Family {hkl}. Negative intercept → bar.

e.g. 1/2, ∞, 1/3 → 2, 0, 3 → (203). Intercepts 1/2, 2/3, −2/3 → 2, 3/2, −3/2 → (4 3 −3)

Line of intersection of two cubic planes = $n_1 \times n_2$ (cross product of hkl's). (110)×(11−1) = [−1 1 0]

Hex 4-index: [UVW]→[uvtw]: u= (2U−V)/3, v=(2V−U)/3, t=−(u+v), w=W. Planes (hkil), i = −(h+k)

Ceramics

r_C/r_A	CN	Geometry
< 0.155	2	linear
0.155–0.225	3	triangular
0.225–0.414	4	tetrahedral
0.414–0.732	6	octahedral
0.732–1.0	8	cubic

Structure	Type	CN C/A	anion pack	n'
Rock salt (NaCl)	AX	6/6	FCC	4
CsCl	AX	8/8	SC	1
Zinc blende (ZnS)	AX	4/4	FCC	4
Fluorite (CaF_2)	AX_2	8/4	SC	4
Antifluorite	A_2X	4/8	FCC	4
Perovskite	ABX_3	12(A),6(B)/6	FCC	1
Spinel	AB_2X_4	4, 6/4	FCC	8

$\rho = n'(\Sigma A_C + \Sigma A_A)/(V_C N_A)$ · n' = formula units/cell = $\rho V_C N_A/(\Sigma A)$

Rock salt a = $2(r_C + r_A)$ · CsCl a = $2(r_C+r_A)/\sqrt{3}$ · zinc blende a = $4(r_C+r_A)/\sqrt{3}$

Ceramic APF: $V_s = n'\cdot[\#cat\cdot(4/3)\pi r_C^3 + \#an\cdot(4/3)\pi r_A^3] / V_C$. Cr_2O_3 (n'=18 hex): 36 Cr + 54 O.

A_2X (antifluorite): X ions at ALL cube corners, HALF of cube centers occupied by A.

Polymers

Polymer	Repeat unit	m (g/mol)
Polyethylene PE	C_2H_4 – CH_2–CH_2–	28.05
Polypropylene PP	C_3H_6 (CH_3 side)	42.08
PVC	C_2H_3Cl (Cl side)	62.50
PTFE	C_2F_4	100.02
Polystyrene PS	C_8H_8 (C_6H_5 side)	104.14
PMMA	$C_5H_8O_2$	100.12
Poly(vinyl alcohol)	C_2H_4O (OH side)	44.05
Isobutylene / isoprene	C_4H_8 / C_5H_8	56.10 / 68.11
Butadiene / styrene	C_4H_6 / C_8H_8	54.09 / 104.14

$DP = \bar{M}_n/m$ · $\bar{M}_n = \Sigma x_i M_i$ · $\bar{M}_w = \Sigma w_i M_i$ (M_i = midpoint of range) · copolymer $\bar{m} = \Sigma f_j m_j$

Chains in a part: mass = ρV → mol chains = mass/(DP·m) → × N_A. Pipe V = $\pi(r_{out}^2 - r_{in}^2)L$

% cryst = $\rho_c(\rho_s - \rho_a) / [\rho_s(\rho_c - \rho_a)] \times 100$

Two samples (C as fraction): $\mathbf{\rho_c = \rho_{s1}\rho_{s2}(C_2 - C_1) / [\rho_{s1}(1-C_1) - \rho_{s2}(1-C_2)]}$

$\mathbf{\rho_a = \rho_{s1}\rho_{s2}(C_1 - C_2) / (C_1\rho_{s1} - C_2\rho_{s2})}$

$\rho_{crystalline} > \rho_{amorphous}$ (tighter packing). Amorphous → transparent; semicrystalline → translucent/opaque.

Crystallizes more: linear > branched > crosslinked > network (≈0) · isotactic ≈ syndiotactic > atactic · simple small side groups · homopolymer > random copolymer. Block vs graft: can't tell.

Thermoplastic (linear/branched): softens, melts, recyclable (PE, PP, PVC, PS). **Thermoset** (crosslinked/network): chars/decomposes, not recyclable (epoxy, phenol-formaldehyde). Elastomer = lightly crosslinked (vulcanized).

T_g: amorphous regions go brittle (below) ↔ rubbery (above). Metals have no T_g. Copolymers: random, alternating, block, graft.

Point defects & composition

$N_v = N \exp(-Q_v/kT)$, $N = N_A\rho/A$ ($\times 10^6$ for $cm^{-3} \rightarrow m^{-3}$). Q_v in eV/atom → $k = 8.62\times10^{-5}$; in J/mol → R.

Fraction = N_v/N. Ratio at T_1, T_2: $\exp[-(Q_v/k)(1/T_1 - 1/T_2)]$. Vacancies ↑ with T.

wt% → at%: $C'_1 = C_1A_2/(C_1A_2 + C_2A_1) \times 100$ · general: $C'_i = (C_i/A_i)/\Sigma(C_j/A_j) \times 100$

at% → wt%: $C_i = C'_iA_i/\Sigma(C'_jA_j) \times 100$

kg/m³: $C''_1 = C_1/(C_1/\rho_1 + C_2/\rho_2) \times 10^3$ (C in wt%, ρ in g/cm³)

Alloy $\rho_{ave} = 100/(C_1/\rho_1 + C_2/\rho_2)$ · $A_{ave} = 100/(C_1/A_1 + C_2/A_2)$

Hume-Rothery (complete solid solubility): (1) Δr < 15% $[(r_{big}-r_{small})/r_{big}]$ (2) same crystal structure (3) similar electronegativity (4) same valence. Most important: (1), (2).

Interstitial site radius r: SC face 0.414R, body 0.732R · FCC oct 0.414R, tet 0.225R · BCC oct 0.155R, tet 0.291R

Schottky: cation + anion vacancies in charge-neutral ratio (NaCl 1+1, Al_2O_3 2 Al + 3 O, $MgCl_2$ 1 Mg + 2 Cl); no interstitials. **Frenkel:** cation leaves site → nearby interstitial (vacancy-interstitial pair). Fewer defects when more ions needed/higher charge/lower T.

Line, area defects

Edge: b ⊥ line (extra half plane) · **Screw:** b ∥ line (helical ramp) · **Mixed:** neither (real crystals). Dislocations move easily in metals, barely in ceramics (directional covalent / charge repulsion in ionic) → brittle. Grain boundaries, surfaces, twin boundaries = 2D. ASTM grain size: $N = 2^{n-1}$ grains/in² at 100×.

Diffusion

Fick 1 (steady): $J = -D\,dC/dx = -D(C_B - C_A)/(x_B - x_A)$ · $J = M/(At) \rightarrow M = JAt$

Position for conc. C: $x = D(C_A - C)/J$. $D = J\cdot\Delta x/\Delta C$ (use wt% → kg/m³ first)

Fick 2 (C_s const): $(C_x - C_0)/(C_s - C_0) = 1 - \text{erf}(z)$, $z = x/(2\sqrt{(Dt)}) \rightarrow t = x^2/(4z^2D)$, $D = x^2/(4z^2t)$

Same conc., same T: $x_1^2/t_1 = x_2^2/t_2$. Same x, different T: $D_1t_1 = D_2t_2$. Diffusion distance ~ $\sqrt{(Dt)}$

Thin film (fixed β, drive-in): $C = [\beta/(2\sqrt{(\pi Dt)})]\exp(-x^2/4Dt)$. Carburizing = C_s const (erf).

$D = D_0\exp(-Q_d/RT)$ · $\ln D = \ln D_0 - (Q_d/R)(1/T)$ · $\log D = \log D_0 - Q_d/(2.3R)\cdot(1/T)$

$T = Q_d/[R(\ln D_0 - \ln D)]$

Two points: $Q_d = -R\ln(D_2/D_1)/(1/T_2 - 1/T_1)$ · $D_0 = D_1\exp(Q_d/RT_1)$

log D vs 1/T plot: slope = $-Q_d/(2.3R)$ → $Q_d = -2.3R\cdot\text{slope}$. Steeper line = higher Q_d.

T for flux J through plate: $D = J\cdot\Delta x/\Delta C$, then $T = Q_d/[R\ln(D_0/D)]$

Interpolate: $z = z_1 + (\text{erf} - \text{erf}_1)(z_2 - z_1)/(\text{erf}_2 - \text{erf}_1)$

z	erf	z	erf	z	erf
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Species in host	D_0 (m²/s)	Q_d (J/mol)
C in α-Fe (BCC)	1.1×10^{-6}	87,400
C in γ-Fe (FCC)	2.3×10^{-5}	148,000
N in α-Fe	5.0×10^{-7}	77,000
N in γ-Fe	9.1×10^{-5}	168,000
Fe in α-Fe	2.8×10^{-4}	251,000
Fe in γ-Fe	5.0×10^{-5}	284,000
Cu in Cu	2.5×10^{-5}	200,000
Al in Al	2.3×10^{-4}	144,000
Mg in Mg	1.5×10^{-4}	136,000
Zn in Zn	1.5×10^{-5}	94,000
Mo in Mo	1.8×10^{-4}	461,000
Ni in Ni	1.9×10^{-4}	285,000
Zn in Cu	2.4×10^{-5}	189,000
Cu in Zn	2.1×10^{-4}	124,000
Cu in Al	6.5×10^{-5}	136,000
Mg in Al	1.2×10^{-4}	130,000
Cu in Ni	2.7×10^{-5}	256,000
Ni in Cu	1.9×10^{-4}	230,000

Faster diffusion: interstitial > vacancy (Q_i lower: no vacancy formation energy, many open sites; C in Fe ≫ Fe self) · grain boundaries/dislocations/surfaces (open, disordered; $Q \approx 0.5Q_{bulk}$, dominate at low T, bulk at high T) · open structures > close-packed (BCC > FCC) · cations > anions (smaller) · secondary bonding > covalent · lower T_m · smaller diffusing atom · higher T

Material selection (guaranteed Q)

1) Read conditions (T range, loads, weight, environment, electrical). 2) Rank key properties. 3) Eliminate (e.g. polymers with T_g in range, too dense, too brittle, conductive). 4) Pick 2–3, compare pros/cons. 5) Defend one + say how to fix its drawback (e.g. Al conductive → insulating coating/standoffs). Space/launch → low ρ, low CTE, wide T range. Any well-argued answer gets credit.